

Distributed Acoustic Sensing Data Processing and Material Classification

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Contents

1	Introduction	2
2	Implementation Process	2
2.1	Data Acquisition and Preprocessing	2
2.2	Channel Selection and Event Extraction	3
2.3	Data Curation and Batch Processing	5
2.4	Classification (k-NN)	5
2.4.1	Euclidean Distance	5
2.4.2	Discussion and Conclusion on k-NN Performance (Euclidean) . . .	6
2.4.3	Dynamic Time Warping (DTW)	6
2.4.4	Discussion and Conclusion on k-NN Performance (DTW)	6
2.4.5	Cross-Correlation	7
2.4.6	Discussion and Conclusion on k-NN Performance (Cross-Correlation)	7
2.4.7	Comparison of the Distance Metrics	7
2.5	Challenges Encountered	8
2.6	Key Findings	8
3	Alignment with the Proposed Plan	9
3.1	Achieving the goals	9
3.2	Lessons Learned and Future Improvements	9
	References	9

1 Introduction

We studied Distributed Acoustic Sensing (DAS) as a powerful technology that uses fiber optic cables to measure pressure and acoustic vibrations along the fiber. The first step was to process raw DAS signals to detect and classify acoustic events occurring in different surrounding materials: *asphalt*, *sand*, and a *technical chamber* at a known distance.

Our main hypothesis was that different different materials would yield distinct signal signatures (e.g., variations in phase, amplitude, and frequency response) when subjected to an impact. After properly filtering the noise and isolating the precise moment of the event, we hypothesized that a machine learning algorithm, specifically a k-Nearest Neighbors (k-NN) classifier, could accurately distinguish the material based on the time-series profile of the acoustic event.

2 Implementation Process

Our pipeline was implemented in several automated stages, bridging raw data acquisition and classification:

2.1 Data Acquisition and Preprocessing

Raw ‘.mat’ data was automatically retrieved from a remote server using a FileZilla FTP setup. The following picture show the RAW waterfall of the fiber data, it is inoperable without any bandpass filtering.

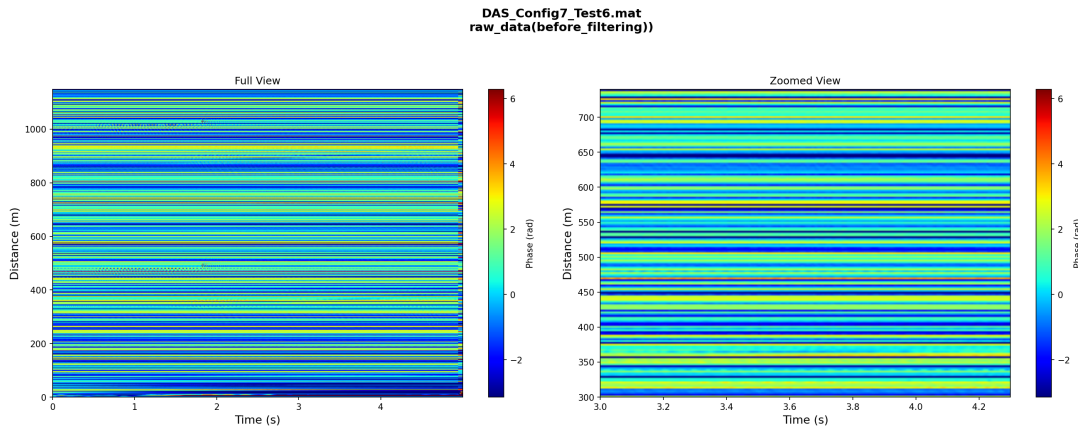


Figure 1: RAW Waterfall plot of DAS dataset

We were provided with an initial `preprocessing.py` script, which we modified heavily. Originally designed to simply plot waterfalls after applying a bandpass filter, we adapted it to output processed differential phase arrays for downstream tasks.

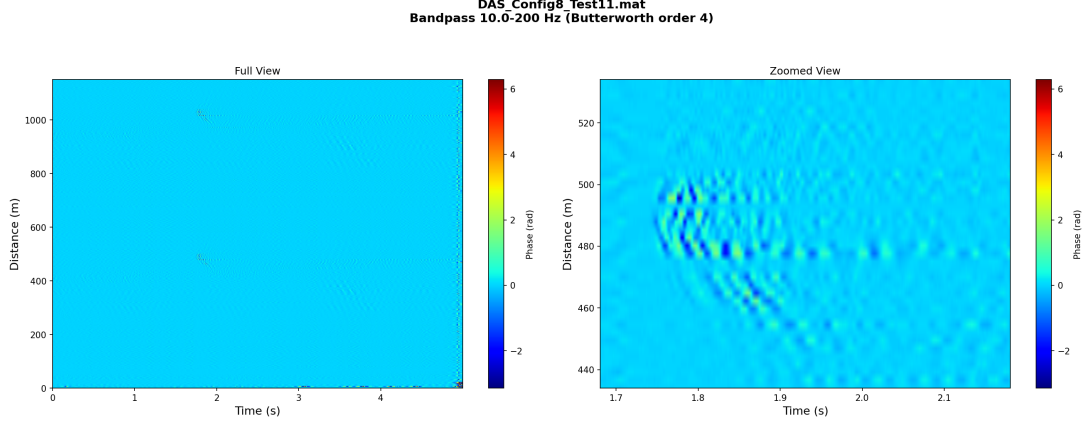


Figure 2: Waterfall plot of DAS after bandwidth filtration

Before applying any signal processing, it was crucial to properly handle the raw baseline data. We implemented a custom parsing logic to directly load the `.mat` files, allowing us to precisely extract the raw differential phase matrix alongside its structural axes (time and spatial resolution).

To establish a reliable baseline, we developed an automated saving routine that exports this raw, unfiltered data into optimized `.npz` files. This function simultaneously generates and saves the initial unfiltered waterfall plots. Having this raw visualization and dataset readily available was essential for comparing the impact of our subsequent filtering techniques.

Following this raw data extraction, we had to apply a filter because raw DAS signals are highly susceptible to low-frequency thermal drift and high-frequency optical noise. The selection of this filter was conducted empirically and visually. We filtered the raw waterfall with different band filters (3-10 Hz, 10-50 Hz, 50-100 Hz, and 100-200 Hz). By generating and visually comparing the waterfall plots (2D arrays of distance versus time) for each band, we observed which frequency ranges best isolated the physical structural waves of the impact. We concluded that a composite Butterworth bandpass filter spanning 10 to 200 Hz (order 4) yielded the cleanest signal morphology.

So we applied a Butterworth bandpass filter (10-200 Hz, order 4), outputting clean 2D arrays of distance versus time.

2.2 Channel Selection and Event Extraction

A major challenge in this phase was bridging the physical realities of the experimental setup with our matrix data structures. We needed to isolate the precise location (channel) and timestamp of the impact. The dataset included three material configurations at specific physical distances along the fiber:

- **Asphalt:** 484 meters
- **Sand:** 146 meters
- **Technical Chamber:** 178 meters

To map these physical distances to specific channel indices in the NumPy array, we had to rely on the physical parameters of the DAS system. The raw spatial resolution of the optical sampling was 0.5 m (So there was a detector every 0.5m). However, because

we applied a spatial downsampling factor (`LOW_RESOL_FACTOR = 5`) to reduce the size of the waterfalls, the effective distance between adjacent channels changed. We calculated the new spatial step size as $\Delta x = \text{spatial_res} \times \text{LOW_RESOL_FACTOR}$ (resulting in 2.5 m). The exact target channel index, r_{search} , could then be reliably determined using $r_{\text{search}} = \lfloor \text{distance} / \Delta x \rfloor$.

Once the correct spatial channel was isolated, we needed to extract a strict 0.5-second time window containing the actual impact event. We experimented with two distinct algorithmic methods:

1. **Sliding Energy Window:** Our initial approach aimed to locate the 0.5-second interval containing the highest total signal energy, defined as the sum of squared amplitudes. A naive implementation using a nested loop would recalculate these sums from scratch at each step, resulting in an $\mathcal{O}(N \times W)$ complexity that is computationally inefficient. To optimize this process to a linear $\mathcal{O}(N)$ complexity, we formulated the problem as a sequence of cumulative moving window sums. By squaring the signal data and computing its cumulative sum using NumPy’s `cumsum` function, the total energy within any window of size W could be found instantaneously through a simple subtraction of two shifted indices. The starting index of the highest-energy event was then identified by searching for the maximum value within this resulting array.
2. **Absolute Maximum Peak:** We implemented an alternative method that searched directly for the maximum absolute peak value in the channel (`np.argmax(np.abs(channel))`) and anchored the extraction window around this distinct spike.

Empirically, the absolute maximum peak method proved to be significantly more robust for our dataset. The sliding energy window was occasionally misled by prolonged background noise sequences that artificially inflated the total energy, whereas the absolute peak reliably locked onto the sharp, distinct impulse of the physical impacts.

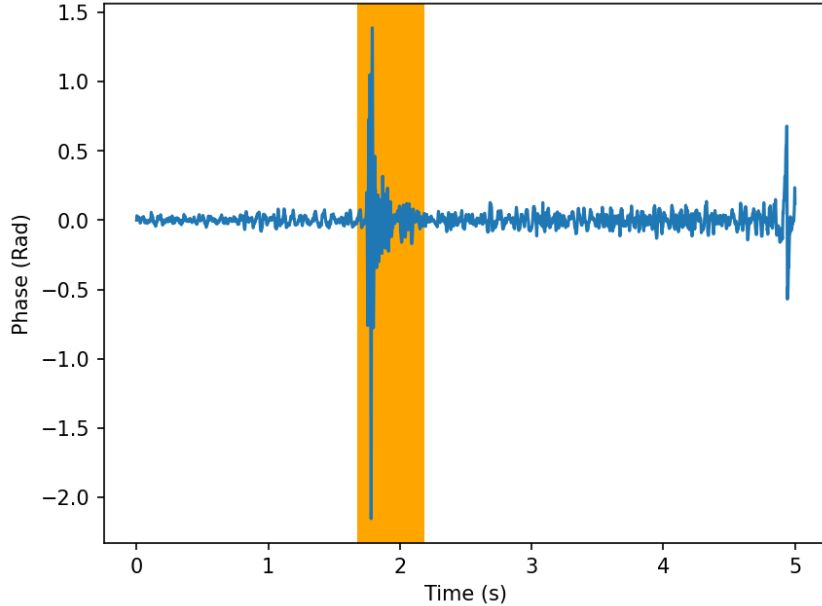


Figure 3: Extraction of the event in the time serie of the event

2.3 Data Curation and Batch Processing

To ensure the quality of our dataset, we plotted the extracted time series and their corresponding cropped waterfalls. We manually verified these plots to discard excessively noisy or corrupted samples. The clean 0.5s time series were then saved as `.npz` files. We streamlined this entirely by creating a mass-extraction script (`programme_extraction_masse.py`) to process all local files automatically.

A pre-step has been made on the raw datasets gave by our supervisor to remove the telltale test of the experiment. Then the program downloaded all raw datasets and automatized the knowledge of the material used during the experiment (in order to know the distance of the researched event).

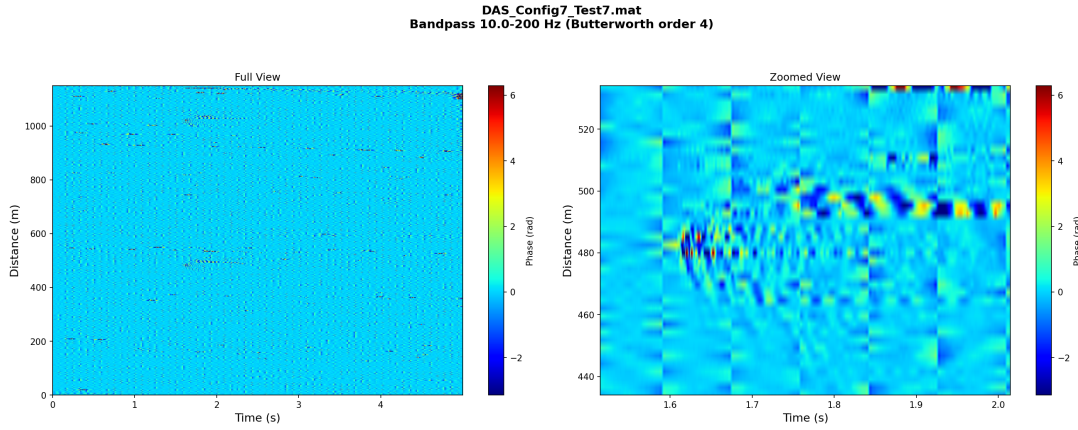


Figure 4: Waterfall of a noisy DAS

2.4 Classification (k-NN)

As the final step in our processing pipeline, we developed a supervised k-NN classifier (implemented in `classify.py`) to automatically predict the ground material based on the previously extracted time series.

The k-NN algorithm is an intuitive and non-parametric method that classifies a new unknown event by comparing its time series to all labeled events in the training set. The predicted class corresponds to the majority class among its k closest neighbors.

Because our inputs are 1D arrays representing acoustic wave propagation over time, the performance of this classifier strongly depends on how the distance, or similarity, between two signals is defined. To determine the most suitable approach for our DAS time series, we implemented and evaluated three distance metrics: Euclidean distance, Dynamic Time Warping (DTW), and Cross-Correlation.

The updated implementation computes the pairwise distance matrix for each metric and evaluates the k-NN classifier using Leave-One-Out cross-validation. Several values of k are tested in order to analyse the influence of the number of neighbours on classification performance.

2.4.1 Euclidean Distance

The Euclidean distance computes the point-to-point difference between two time series:

- **Pros:** It is extremely fast to compute, with a complexity of $\mathcal{O}(N)$, making it highly scalable for large datasets. It is also mathematically straightforward and easy to implement.
- **Cons:** Its main limitation is its sensitivity to temporal shifts. A small delay between two otherwise similar signals can significantly increase the distance because corresponding points are no longer aligned. Euclidean distance is also unable to handle local temporal distortions, such as stretching or compression of a waveform.

2.4.2 Discussion and Conclusion on k-NN Performance (Euclidean)

Using Euclidean distance, the k-NN classifier achieves an accuracy of approximately 0.73 (73%) on the updated dataset.

This result highlights the limitations of point-to-point comparison for DAS acoustic signals. Euclidean distance assumes that two signals are perfectly aligned in time and compares each sample at a fixed position with the corresponding sample of the other time series. Consequently, even a small temporal shift can lead to a significant increase in the measured distance.

In the context of acoustic wave propagation, signals generated by the same ground material may present slight differences in propagation time or waveform alignment. These variations can therefore make two physically similar signals appear more distant according to the Euclidean metric.

Although Euclidean distance remains attractive because of its simplicity and low computational cost, its classification accuracy of 73% is the lowest among the three approaches evaluated. This suggests that temporal alignment is an important factor for accurately classifying the considered DAS time series or that the event extraction creates an offset in time.

2.4.3 Dynamic Time Warping (DTW)

Unlike the rigid Euclidean distance, Dynamic Time Warping creates an elastic mapping between two time series by warping the time axis in order to find an optimal alignment between their shapes.

- **Pros:** DTW is highly robust to temporal misalignment because it does not require an exact point-to-point correspondence. It can therefore identify similar patterns even when a signal is locally stretched or compressed.
- **Cons:** The computation is significantly more expensive than Euclidean distance, with a typical complexity of $\mathcal{O}(N^2)$ for comparing two time series. This can become a significant bottleneck when a large number of pairwise distances must be computed. In addition, without appropriate constraints, DTW may over-warp signals and potentially align noise with meaningful features.

2.4.4 Discussion and Conclusion on k-NN Performance (DTW)

Using Dynamic Time Warping, the k-NN classifier achieves an accuracy of approximately 0.87 (87%), which is a clear improvement compared with the 73% obtained using Euclidean distance.

The improved performance can be explained by the ability of DTW to compensate for temporal distortions between signals. Instead of enforcing a rigid point-to-point alignment, DTW finds an elastic correspondence between the two waveforms. This allows signals with similar structures to remain close even when parts of their waveforms are locally shifted, stretched, or compressed.

The results therefore confirm that temporal alignment is important for the considered DAS signals. DTW provides greater robustness than Euclidean distance, although this improvement comes at a significantly higher computational cost.

2.4.5 Cross-Correlation

Cross-Correlation compares two signals over a range of temporal shifts. In our implementation, the signals are first normalized before computing the correlation, and the maximum absolute correlation value is used to define the distance.

- **Pros:** Cross-Correlation is robust to global temporal shifts because it searches for the delay at which the two signals are most strongly aligned. The normalization also reduces the influence of amplitude differences and makes the comparison focus more strongly on waveform shape.
- **Cons:** Cross-Correlation does not provide the same local temporal flexibility as DTW. It can compensate for a global shift, but it cannot explicitly correct local stretching or compression of a signal. Its computational cost is also higher than that of Euclidean distance.

2.4.6 Discussion and Conclusion on k-NN Performance (Cross-Correlation)

Among the three distance metrics evaluated, Cross-Correlation provides the best classification performance, achieving an accuracy of approximately 0.90 (90%) on the updated dataset.

This result suggests that the acoustic signals corresponding to the different ground materials share characteristic waveform structures, while also exhibiting temporal shifts between measurements. By searching for the best temporal alignment, Cross-Correlation can compare the morphology of the signals while reducing the influence of these global delays.

The normalization applied before computing the correlation further reduces the sensitivity to amplitude variations. This is particularly relevant for DAS measurements, where differences in signal amplitude do not necessarily correspond to differences in the underlying ground material.

Cross-Correlation slightly outperforms DTW, with an accuracy of 90% compared with 87%. This indicates that, for the dataset considered in this study, compensating for global temporal shifts is sufficient to capture most of the relevant similarities between the signals. While DTW provides greater flexibility for local temporal distortions, this additional flexibility does not result in a higher classification accuracy on the current dataset.

2.4.7 Comparison of the Distance Metrics

The classification results obtained with the three distance metrics are summarized below:

Distance Metric	Accuracy
Euclidean	73%
DTW	87%
Cross-Correlation	90%

Table 1: Comparison of k-NN classification accuracy for the three distance metrics.

The results clearly show that the choice of distance metric has a significant impact on the performance of the k-NN classifier. Euclidean distance obtains the lowest accuracy (73%), mainly because of its sensitivity to temporal misalignment. DTW improves the classification performance to 87% by allowing elastic temporal alignment, while Cross-Correlation achieves the best result with an accuracy of 90%.

These results indicate that the temporal structure of the acoustic waveforms contains relevant information for distinguishing the different ground materials. In particular, the improvement obtained with DTW and Cross-Correlation compared with Euclidean distance demonstrates that temporal alignment is an important factor in the classification problem.

Overall, Cross-Correlation provides the best classification performance among the three tested approaches. It offers a good compromise between robustness to temporal shifts and computational complexity, making it the most promising distance metric for the current DAS material classification pipeline. DTW also provides strong performance, but its higher computational cost makes Cross-Correlation more attractive for a larger-scale implementation. Euclidean distance, although computationally efficient, appears less suitable for the considered acoustic time series due to its lack of temporal alignment robustness.

2.5 Challenges Encountered

- **Bridging Physics and Code:** one of the most significant barrier was to understand the physical parameters of the DAS system (such as optical sampling rate, gauge length, and spatial resolution) and translating them into code to locate the correct channel index on the fiber.
- **Event Alignment:** Initially, isolating the exact 0.5s window of the impact was difficult. Background noise often threw off our sliding-window energy approach.

2.6 Key Findings

- **Extraction Method:** The maximum absolute peak method outperformed the sliding energy window for event localization.
- **k-NN Distance Metrics:** Our classification results revealed that **Cross-Correlation** yielded the highest accuracy (reaching 90%), demonstrating excellent robustness to global time shifts.
- **Metric Comparison:** **DTW** remains theoretically accurate but proved to be extremely time-consuming with severe computational bottlenecks. Conversely, **Euclidean distance** was the fastest to compute but the most sensitive to misalignments, yielding a lower accuracy of 73% due to its inability to account for phase shifts in the wave signals.

3 Alignment with the Proposed Plan

3.1 Achieving the goals

From our point of view we successfully completed the pipeline from raw data parsing to a successful classification by k-NN. We managed to automate the mass extraction of events, which was highly beneficial for iterating on the machine learning models with all files extracted. Manual sorting ensured that our classifier was trained on valid data without any noised datasets, though it required human intervention.

3.2 Lessons Learned and Future Improvements

If we were to start over, we would implement an automated outlier detection method (e.g., using signal-to-noise ratio thresholds) to replace the manual visual verification of the waterfalls. This would make the pipeline 100% autonomous.

Furthermore, we learned that complex algorithms like DTW, while theoretically sound for time series, are often impractical without extensive optimization compared to efficient alternatives like Cross-Correlation.

Finally, the automated ultimate objective could be to pre-train an AI tool able to get over time and distance the fiber warping in live. It should at least, be able to find events across an input fiber signal, extract there time series and be able to classify it from learned labels.

References

- [1] P. Pruvost, É. Awwad, H. Huang, and Y. Jaouën, “Frequency-Domain Characterization of Deployed Fiber-Medium Coupling using DAS,” in *Proceedings of the Optical Fiber Communication Conference (OFC) 2026*, Optica Publishing Group, 2026, paper Th2A.59.